

Name: _____

Date: _____

SURDS, (SIMPLIFYING SURDS)

LO: I can simplify surds by finding the largest square factor.

Simplifying surds

To **simplify a surd**, find the largest **perfect square factor** of the number under the root sign and take its square root outside.

Method: $n = (a^2 \times b) = a \sqrt{b}$. Choose the largest perfect square factor for efficiency.

Quick examples

●	$12 = (4 \times 3) = 2 \sqrt{3}$	4 is the square factor
●	$50 = (25 \times 2) = 5 \sqrt{2}$	25 is the largest square factor
●	$72 = (36 \times 2) = 6 \sqrt{2}$	36 is the largest square factor
●	$12 = 6 \times 2$	not simplified; need square factors
●	$50 = (10 \times 5) = 10 \sqrt{5}$	neither factor is a perfect square

Key idea

Find the largest perfect square that divides the number.
Take its root outside the radical.

$$n = \textcircled{a} \textcircled{b} \text{ where } n = a^2 \times b$$

b should have no perfect square factors left

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – basic simplification

Simplify 20 .

Find largest square factor of 20: 4

$$20 = (4 \times 5) \\ = 2 \sqrt{5}$$

2 5

4 is the largest perfect square that divides 20.

Example 2 – larger number

Simplify 45 .

Find largest square factor of 45: 9

$$45 = (9 \times 5) \\ = 3 \sqrt{5}$$

3 5

9 is the largest perfect square that divides 45.

Example 3 – large radicand

Simplify 200 .

Find largest square factor of 200:

$$100 \\ 200 = (100 \times 2) \\ = 10 \sqrt{2}$$

10 2

Look for 4, 9, 16, 25, 36, 49, 64, 81, 100 as factors.

Example 4 – with coefficient

Simplify $3 \sqrt{28}$.

$$28 = (4 \times 7) = 2 \sqrt{7} \\ 3 \times 2 \sqrt{7} = 6 \sqrt{7}$$

Multiply coefficients

6 7

Simplify the surd first, then multiply by the outer coefficient.

YOUR TURN – PRACTICE (deliberate practice)

1. Basic simplification

Simplify each surd.

- a) 8
- b) 18
- c) 32
- d) 27

2. Larger numbers

Simplify each surd.

- a) 48
- b) 75
- c) 98
- d) 125

3. With coefficients

Simplify fully.

- a) 2 12
- b) 5 8
- c) 3 50
- d) 4 45

4. Expressing in surd form

Write in the form $a\sqrt{b}$.

- a) 180
- b) 128
- c) 300
- d) 242

5. Mixed surd simplification

Simplify each expression fully.

- a) 63
- b) $2\sqrt{54}$
- c) 108
- d) $72\sqrt{2}$
- e) 500

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $12 = 2\sqrt{3}$ because $12 = 4 \times 3$ ☐

Correct answer: _____

Reason: _____

b) $50 = 10\sqrt{2}$ ☐

Correct answer: _____

Reason: _____

c) $72 = 6\sqrt{2}$ because $72 = 36 \times 2$ ☐

Correct answer: _____

Reason: _____

d) $8 = 4\sqrt{2}$ ☐

Correct answer: _____

Reason: _____

e) A surd is fully simplified when no perfect square divides the radicand ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Find the exact length of the diagonal of a rectangle with sides **6 cm** and **8 cm**. Express 100 and show it equals a whole number. Now find the diagonal of a **5 cm by 7 cm** rectangle in simplified surd form.

Expression: _____

Simplified: _____

b) Simplify $(3n^2)$ where n is a positive integer. Hence write **(300)** in the form $a\sqrt{b}$ and verify your answer by finding $a^2 \times b$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

